

Postulates and Theorems

Euclid's Postulates

POSTULATE EUC-1

Any two points can be connected with a line segment.

POSTULATE EUC-2

Any line segment can be extended indefinitely in two directions to make a line.

POSTULATE EUC-3

Given any line segment, a circle can be drawn that has the segment as its radius and one endpoint as its center.

POSTULATE EUC-4

All right angles are equivalent to each other.

POSTULATE EUC-5

Given any straight line and a point not on the line, there is exactly one line through the point that is parallel to the line.

General

POSTULATE GEN-1

Two points determine a line.

POSTULATE GEN-2

Three noncollinear points determine a plane.

Measurement

POSTULATE MEAS-1

Ruler Postulate The points on a line can be numbered so that positive number differences measure distances.

POSTULATE MEAS-2

Segment Addition Postulate If B is between A and C , then $AB + BC = AC$. Also, if $AB + BC = AC$ and A , B , and C are collinear, then B is between A and C .

POSTULATE MEAS-3

Segment Congruence Postulate If two segments have the same length as measured by a fair ruler, then the segments are congruent ($\cong$). Also, if two segments are congruent, then they have the same length as measured by a fair ruler.

POSTULATE MEAS-4

Angle Addition Postulate If point D lies in the interior of $\angle ABC$, then $m\angle ABD + m\angle DBC = m\angle ABC$.

POSTULATE MEAS-5

Angle Congruence Postulate If two angles have the same measure as measured by a protractor, then the angles are congruent. Also, if two angles are congruent, then they have the same measure as measured by a protractor.

POSTULATE MEAS-6

Linear Pair Postulate If two angles form a linear pair, then they are supplementary angles.

THEOREM MEAS-1

Vertical Angles Theorem If two angles form a pair of vertical angles, then they are congruent.

Parallels

POSTULATE PAR-1

Corresponding Angles Postulate If two parallel lines are intersected by a transversal, then corresponding angles are congruent.

POSTULATE PAR-2

Converse of the Corresponding Angles Postulate If two coplanar lines are intersected by a transversal and the corresponding angles are congruent, then the lines are parallel.

THEOREM PAR-1

Alternate Interior Angles Theorem If two parallel lines are intersected by a transversal, then the alternate interior angles are congruent.

THEOREM PAR-2

Alternate Exterior Angles Theorem If two parallel lines are intersected by a transversal, then the alternate exterior angles are congruent.

THEOREM PAR-3

Same-Side Interior Angles Theorem If two parallel lines are intersected by a transversal, then the same-side interior angles are supplementary.

THEOREM PAR-4

Converse of the Alternate Interior Angles Theorem If two coplanar lines are intersected by a transversal and the alternate interior angles are congruent, then the lines are parallel.

THEOREM PAR-5

Converse of the Alternate Exterior Angles Theorem If two coplanar lines are intersected by a transversal and the alternate exterior angles are congruent, then the lines are parallel.

THEOREM PAR-6

Converse of the Same-Side Interior Angles Theorem If two coplanar lines are intersected by a transversal and the same-side interior angles are supplementary, then the lines are parallel.

Lines

POSTULATE LINES-1

Parallel Postulate Given a line and a point not on the line, there is one and only one line that contains the given point and is parallel to the given line.

THEOREM LINES-1

If two coplanar lines are perpendicular to ($\perp$) the same line, then the two lines are parallel.

THEOREM LINES-2

If two coplanar lines are parallel to the same line, then the two lines are parallel.

Coordinate

THEOREM COORD-1

Parallel Lines Theorem Two coplanar nonvertical lines are parallel if and only if they have the same slope. Any two vertical lines are parallel.

THEOREM COORD-2

Perpendicular Lines Theorem Two coplanar nonvertical lines are perpendicular if and only if the product of their slopes equals -1 . Any vertical line is perpendicular to any horizontal line.

Congruence

POSTULATE CONG-1

Polygon Congruence Postulate Two polygons are congruent if and only if there is a correspondence between their sides and angles so that all pairs of corresponding angles are congruent and all pairs of corresponding sides are congruent.

POSTULATE CONG-2

Side-Side-Side (SSS) Congruence Postulate If the three sides of one triangle are congruent to the three sides of another triangle, then the two triangles are congruent.

POSTULATE CONG-3

Side-Angle-Side (SAS) Congruence Postulate If two sides and the included angle in one triangle are congruent to two sides and the included angle in another triangle, then the two triangles are congruent.

POSTULATE CONG-4

Angle-Side-Angle (ASA) Congruence Postulate If two angles and the included side in one triangle are congruent to two angles and the included side in another triangle, then the two triangles are congruent.

THEOREM CONG-1

Hypotenuse-Leg (HL) Congruence Theorem If the hypotenuse and a leg of one right triangle are congruent to the hypotenuse and corresponding leg of another right triangle, then the two triangles are congruent.

Triangles

THEOREM TRI-1

Triangle Sum Theorem The sum of the measures of the interior angles of a triangle is 180° .

THEOREM TRI-2

Exterior Angle Theorem The measure of an exterior angle of a triangle is equal to the sum of the measures of the remote interior angles.

THEOREM TRI-3

Isosceles Triangle Theorem If two sides of a triangle are congruent, then the angles opposite those sides are congruent.

THEOREM TRI-4

Converse of the Isosceles Triangle Theorem If two angles of a triangle are congruent, then the sides opposite those angles are congruent.

Quadrilaterals

THEOREM QUAD-1

In a parallelogram, the opposite sides are congruent.

THEOREM QUAD-2

In a rectangle, the diagonals are congruent.

THEOREM QUAD-3

If two pairs of opposite sides of a quadrilateral are congruent, then the quadrilateral is a parallelogram.

THEOREM QUAD-4

If two opposite sides of a quadrilateral are parallel and congruent, then the quadrilateral is a parallelogram.

THEOREM QUAD-5

If the diagonals of a quadrilateral bisect each other, then the quadrilateral is a parallelogram.

THEOREM QUAD-6

If the diagonals of a parallelogram are congruent, then the parallelogram is a rectangle.

THEOREM QUAD-7

If the diagonals of a parallelogram are perpendicular, then the parallelogram is a rhombus.

THEOREM QUAD-8

If two adjacent sides of a parallelogram are congruent, then the parallelogram is a rhombus.

THEOREM QUAD-9

If the diagonals of a parallelogram bisect the angles of the parallelogram, then the parallelogram is a rhombus.

THEOREM QUAD-10

If one angle of a parallelogram is a right angle, then the parallelogram is a rectangle.

Right Triangles

THEOREM RIGHT-1

Pythagorean Theorem For all right triangles, the square of the length of the hypotenuse c equals the sum of the squares of the lengths of the legs a and b .

$$c^2 = a^2 + b^2$$

THEOREM RIGHT-2

Converse of the Pythagorean Theorem If the square of the length of the longest side of a triangle equals the sum of the squares of the lengths of the other two sides, then the triangle is a right triangle.

THEOREM RIGHT-3

45°-45°-90° Triangle Theorem In any 45°-45°-90° triangle, the length of the hypotenuse is $\sqrt{2}$ times the length of a leg.

THEOREM RIGHT-4

30°-60°-90° Triangle Theorem In any 30°-60°-90° triangle, the length of the hypotenuse is 2 times the length of the shorter leg, and the length of the longer leg is $\sqrt{3}$ times the length of the shorter leg.

Similarity

POSTULATE SIM-1

Polygon Similarity Postulate Two polygons are similar if and only if there is a correspondence between their angles and their sides so that all corresponding angles are congruent and all corresponding sides are proportional.

POSTULATE SIM-2

Angle-Angle (AA) Similarity Postulate If two angles of a triangle are congruent to two angles of another triangle, then the triangles are similar.

COROLLARY SIM-1

Two-Transversal Proportionality Corollary Three or more parallel lines divide two intersecting transversals proportionally.

THEOREM SIM-1

Side-Side-Side (SSS) Similarity Theorem If the three sides of a triangle are proportional to the three sides of another triangle, then the triangles are similar.

THEOREM SIM-2

Side-Angle-Side (SAS) Similarity Theorem If two sides of a triangle are proportional to two sides of another triangle and if their included angles are congruent, then the triangles are similar.

THEOREM SIM-3

Triangle Proportionality Theorem A line parallel to one side of a triangle divides the other two sides proportionally.

THEOREM SIM-4

Angle Bisector Theorem An angle bisector of an angle of a triangle divides the opposite side in two segments that are proportional to the other two sides of the triangle.

Circles

COROLLARY CIRC-1

Right-Angle Corollary An angle that is inscribed in a semicircle is a right angle.

COROLLARY CIRC-2

Arc-Intercept Corollary Two inscribed angles that intercept the same arc have the same measure.

COROLLARY CIRC-3

Converse of the Right-Angle Corollary If an inscribed angle is a right angle, then the intercepted arc is a semicircle.

THEOREM CIRC-1

Chords and Arcs Theorem In a circle or in congruent circles, the arcs of congruent chords are congruent.

THEOREM CIRC-2

Converse of the Chords and Arcs Theorem In a circle or in congruent circles, the chords of congruent arcs are congruent.

THEOREM CIRC-3

Tangent Theorem A line that is tangent to a circle is perpendicular to a radius of the circle at the point of tangency.

THEOREM CIRC-4

Converse of the Tangent Theorem A line that is perpendicular to a radius of a circle at its endpoint on the circle is tangent to the circle.

THEOREM CIRC-5

Radius and Chord Theorem A radius that is perpendicular to a chord of a circle bisects the chord.

THEOREM CIRC-6

Inscribed Angle Theorem An angle inscribed in a circle has a measure that equals one-half the measure of its intercepted arc.

THEOREM CIRC-7

Inscribed Quadrilateral Theorem If a quadrilateral is inscribed in a semicircle, then the opposite angles are supplementary.

THEOREM CIRC-8

If a tangent and a secant or chord intersect on a circle at the point of tangency, then the measure of the angle formed equals one-half the measure of the intercepted arc.

THEOREM CIRC-9

The measure of a secant-tangent angle with its vertex outside the circle equals one-half of the difference of the measures of the intercepted arcs.

THEOREM CIRC-10

The measure of an angle that is formed by two secants that intersect in the exterior of a circle equals one-half of the difference of the measures of the intercepted arcs.

THEOREM CIRC-11

The measure of a tangent-tangent angle with its vertex outside the circle equals one-half of the difference of the measures of the intercepted arcs, or the measure of the major arc minus 180° .

THEOREM CIRC-12

The measure of an angle that is formed by two secants or chords that intersect in the interior of a circle equals one-half the sum of the measures of the arcs intercepted by the angle and its vertical angle.